

NAUFAL ZAIDAN NABHAN

☎ +821033143246 | ✉ naufalz6@skku.edu | 📄 Naufal Zaidan Nabhan

RESEARCH INTERESTS

AI Communications, Wireless Communication Systems, Internet of Things

EDUCATION

Sungkyunkwan University (SKKU)

Doctor of Philosophy

Electrical and Computer Engineering; Full-ride scholarship from STEM Scholarship SKKU

Suwon, South Korea

Mar '26 - Present

Bandung Institute of Technology (ITB)

Master of Science (5-year Integrated Bachelor's and Master's Program)

Electrical Engineering; Full-ride scholarship from SAGATA ITB

Bandung, Indonesia

Jul '23 - Aug '24

Bandung Institute of Technology (ITB)

Bachelor of Science

Telecommunication Engineering

Bandung, Indonesia

Aug '19 - Jul '23

SELECTED PUBLICATIONS

Microwave Wireless Power Transfer for Mid-Air UAV Charging Using a Distributed Tiled Rectenna Architecture in 5G-Enabled Aerial Networks 2026

Joko Suryana, Naufal Zaidan Nabhan, Ali Rhomadoni, M. Iqbal Arrachman, Reuben T.C. Simatupang
IEEE Access

LoRa Based Underwater Wireless Network System

2024

Naufal Zaidan Nabhan, Joko Suryana

2024 10th International Conference on Wireless and Telematics (ICWT '24)

Optimum design of 60WAlGaIn/GaN Broadband RF Amplifier for S-Band Operation

2022

Naufal Zaidan Nabhan, Basuki Rachmatul Alam

2022 International Symposium on Electronics and Smart Devices (ISESD '22)

RESEARCH EXPERIENCES

Radio and Microwave Lab, Telecommunication Engineering Department

Oct '23 - Aug '24

Research Assistant

- Participated in a national multi-institutional consortium project for Ground Controlled Interceptor (GCI) Radar under the supervision of **Dr. Joko Suryana of Bandung Institute of Technology (ITB)**
- Gained hands-on experience in Digital Signal Processing (DSP) techniques for transmitting and processing **Linear Frequency Modulated (LFM)** and **Frequency Modulated Continuous Wave (FMCW)** radar signals, including signal generation and receiver processing
- Designed and developed **LFM** and **FMCW** radar system prototype simulated by **MATLAB** and implemented by **USRP** for short and long moving range

Teaching Factory Manufacturing of Electronics (TFME) Lab

Jun '22 - Aug '22

Research Assistant

- Collaborated with **Dr. Ir. Basuki Rachmatul Alam of Bandung Institute of Technology (ITB)** to work on a research funded by the Indonesian Ministry of Education
- Developed an optimal 60 W RF Power Amplifier for S-band operation, implementing a **chip-based GaN HEMT transistor** wirebonded onto a patch-based circuit on a Rogers Duroid substrate, while **incorporating a bias tee and an impedance matching network**
- Successfully achieved **56%** RF Power Amplifier efficiency

Telematics Lab, Telecommunication Engineering Department

Mar '23 - Aug '24

Research Assistant

- Collaborated under the supervision of **Ir. Wervyan Shalannanda, S.T, M.T of Bandung Institute of Technology (ITB)** to design an E2E early seizure detection system and perform analysis on data transmission
- Designed an early seizure detection system utilizing **EEG electrode cap, OpenBCI Cyton Board, and Raspberry Pi** and **wrote code** to integrate the board with Raspberry Pi
- Analyzed the **E2E data transmission delay** of the system under 3 network conditions (LAN, 2 Subnets, Cloud)

TEACHING

Teaching Assistant	Electromagnetics I & II
Lab. Assistant Coordinator	Communication Systems I & II, Signal Processing

INDUSTRY EXPERIENCES

Radio Network Optimization Engineer, Huawei	Sep '24 - Sep '26
- Support 5G deployment and ensure radio network quality in Indonesia big cities (Jakarta, Bandung, Surabaya, Medan, Semarang) to reach customer satisfaction - Develop a Python-based Root Cause Analysis tools to audit bad network in a certain area which accelerate radio access network assessment and reduce service delivery time.	

TECHNICAL SKILLS

Programming & OS	Python, Linux Kernel, Ubuntu
Internet of Things (IoT)	Raspberry Pi, OpenBCI Cyton, Arduino UNO, ESP8266, ESP32
Electromagnetics Simulation	CST Studio, HFSS
Signal Processing	LabView, MATLAB
Other	MySQL, LaTeX, Tableau

ACTIVITIES

Aquifera Community Service	Aug '23 - Aug '24
- Proposed Waterbox, an IoT based water measure system utilizing ESP32 and MQTT as communication protocol to be deployed in a remote area - Held responsibility for integrating ESP32 and waterflow sensor to PCB circuit, ensuring connectivity of ESP32 and sensors, and validating the functionality of the hardware used	
Bengkel Radio IMT "Signum" ITB Research Project Group	Mar '22 - Feb '23
- Collaborated within a team of 5 members to create a Rogers Duroid substrate Bandstop Filter aimed to attenuate signal frequencies within the range of 3.2 – 3.7 GHz Conducted simulations using specialized software and performed measurements on the fabricated filter using a Vector Network Analyzer (VNA)	

AWARDS

2 nd Winner - FORDIGI BUMN Bandung Chapter Hackathon - BUMN Digital Forum (FORDIGI)	2023
- Participated as a 4 members team in a competitive Hackaton held by the ministry of state-owned enterprises - Proposed a product idea named "EZValve" as an IoT-based system irrigation to simplify farmers work - Won 2nd Winner out of 1840 teams	

HONORS

Bachelor's Scholarship Awardee - ITB Electrical Engineering Alumni Association	2022
Master's Scholarship Awardee - SAGATA Tuition Scholarship, ITB	2023
PhD's Scholarship Awardee - STEM Scholarship, SKKU	2026

REFERENCES

Kae-Won Choi Professor Sungkyunkwan University kaewonchoi@skku.edu	Joko Suryana Associate Professor Bandung Institute of Technology joko@itb.ac.id	Basuki R. Alam Associate Professor Bandung Institute of Technology basuki.rachmatul@stei.itb.ac.id
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